

CS229
Problem Set 3

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Problem 1

- (a) The network implies

$$h_2^{(i)} = g\left((w_{2,1}^{[1]})^T x^{(i)} + w_{0,2}^{[1]}\right), \quad o^{(i)} = g\left((w_2^{[2]})^T h^{(i)} + w_0^{[2]}\right), \quad \text{where } g(z) = \frac{1}{1 + \exp(-z)}$$

The derivative of loss function w.r.t. $w_{1,2}^{[1]}$ is

$$\begin{aligned} \frac{\partial \ell}{\partial w_{1,2}^{[1]}} &= \frac{\partial \ell}{\partial o} \cdot \frac{\partial o}{\partial h_2} \cdot \frac{\partial h_2}{\partial w_{1,2}^{[1]}} \\ &= \frac{2}{m} \sum_{i=1}^m (o^{(i)} - y^{(i)}) \cdot o^{(i)} (1 - o^{(i)}) w_2^{[2]} \cdot h_2^{(i)} (1 - h_2^{(i)}) x^{(i)} \end{aligned}$$

Hence, the GD update is

$$\begin{aligned} w_{1,2}^{[1]} &:= w_{1,2}^{[1]} - \alpha \frac{\partial \ell}{\partial w_{1,2}^{[1]}} \\ &= w_{1,2}^{[1]} - \alpha \cdot \frac{2}{m} \sum_{i=1}^m (o^{(i)} - y^{(i)}) o^{(i)} (1 - o^{(i)}) w_2^{[2]} h_2^{(i)} (1 - h_2^{(i)}) x^{(i)} \end{aligned}$$

- (b) The two classes are seperable by a triangle. The constraint indicating that a data point lies inside of one side of the triangle can be written as

$$w_{1,j}^{[1]} x_1^{(i)} + w_{2,j}^{[1]} x_2^{(i)} + w_{0,j}^{[1]} \geq 0$$

That is to say, if weights and bias are rightly set, given a data point is inside of the triangle,

$$h_j = f(w_{1,j}^{[1]} x_1^{(i)} + w_{2,j}^{[1]} x_2^{(i)} + w_{0,j}^{[1]}) = 1$$

Hence, to achieve 100% accuracy is to ensure $o = 0$ only if all 3 neurons in the hidden layer to output 1 namely, the output neuron is something like

$$o = f(2.5 - h_1 - h_2 - h_3),$$

so that $o = 0$ if and only if the data point is inside of all three sides of the triangle.

A set of weights that can ensure 100% accuracy is

$$\begin{aligned} w_{0,1}^{[1]} &= -0.5, & w_{1,1}^{[1]} &= 1, & w_{2,1}^{[1]} &= 0, \\ w_{0,2}^{[1]} &= -0.5, & w_{1,2}^{[1]} &= 0, & w_{2,2}^{[1]} &= 1, \\ w_{0,3}^{[1]} &= 4, & w_{1,3}^{[1]} &= -1, & w_{2,3}^{[1]} &= -1, \\ w_0^{[2]} &= 2.5, & w_1^{[2]} &= -1, & w_2^{[2]} &= -1, & w_3^{[2]} &= -1. \end{aligned}$$

- (c) Impossible to have 100% accuracy. With the linear activation functions $g(x) = x$ for h_1, h_2 , and h_3 , and step function $f(x)$ for o , the output is simply

$$\begin{aligned} o &= f(w_0^{[2]} + \sum_{j=1}^3 w_j^{[2]} h_j) \\ &= f(w_0^{[2]} + \sum_{j=1}^3 w_j^{[2]} (w_{0,j}^{[1]} + \sum_{i=1}^2 w_{i,j}^{[1]} x_i)) \\ &= f(w_0 + w_1 x_1 + w_2 x_2) \end{aligned}$$

The neural network reduces to a linear classifier. However, the given dataset is not linear-separable, so it is impossible to have 100% accuracy.

Problem 2

(a)

$$\begin{aligned}
 D_{\text{KL}}(P\|Q) &= \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)} \\
 &= \mathbb{E}_{X \sim P} \left[\log \frac{P(X)}{Q(X)} \right] \\
 &= \mathbb{E}_{X \sim P} \left[-\log \frac{Q(X)}{P(X)} \right]
 \end{aligned}$$

Since $f(x) = -\log x$ is convex, according to Jensen's inequality,

$$\begin{aligned}
 D_{\text{KL}}(P\|Q) &= \mathbb{E}_{X \sim P} \left[-\log \frac{Q(X)}{P(X)} \right] \geq -\log \mathbb{E}_{X \sim P} \left[\frac{Q(X)}{P(X)} \right] \\
 &= -\log \sum_{x \in \mathcal{X}} Q(x) \\
 &= -\log 1 \\
 &= 0
 \end{aligned}$$

Note that the inequality has $\mathbb{E}_{X \sim P} \left[-\log \frac{Q(X)}{P(X)} \right] = -\log \mathbb{E}_{X \sim P} \left[\frac{Q(X)}{P(X)} \right]$ if and only if

$$\frac{Q(X)}{P(X)} = \mathbb{E}_{X \sim P} \left[\frac{Q(X)}{P(X)} \right] = 1,$$

i.e. $Q = P$. Hence,

$$\forall P, Q \quad D_{\text{KL}}(P\|Q) \geq 0 \text{ and } D_{\text{KL}}(P\|Q) = 0 \text{ iff } P = Q$$

(b)

$$\begin{aligned}
 D_{\text{KL}}(P(X, Y)\|Q(X, Y)) &= \sum_x \sum_y P(x, y) \log \frac{P(x, y)}{Q(x, y)} \\
 &= \sum_x \sum_y P(y|x)P(x) \log \frac{P(y|x)P(x)}{Q(y|x)Q(x)} \\
 &= \sum_x P(x) \sum_y P(y|x) \left(\log \frac{P(y|x)}{Q(y|x)} + \log \frac{P(x)}{Q(x)} \right) \\
 &= \sum_x P(x) \sum_y P(y|x) \log \frac{P(y|x)}{Q(y|x)} + \sum_x P(x) \log \frac{P(x)}{Q(x)} \sum_y P(y|x) \\
 &= D_{\text{KL}}(P(Y|X)\|Q(Y|X)) + D_{\text{KL}}(P(X)\|Q(X))
 \end{aligned}$$

(c)

$$\begin{aligned}
 D_{\text{KL}}(\hat{P}\|P_\theta) &= \sum_{i=1}^m \hat{P}(x^{(i)}) \log \frac{\hat{P}(x^{(i)})}{P_\theta(x^{(i)})} \\
 &= H(\hat{P}) - \sum_{i=1}^m \hat{P}(x^{(i)}) \log P_\theta(x^{(i)}) \\
 &= H(\hat{P}) - \sum_{i=1}^m \log P_\theta(x^{(i)}) \cdot \frac{1}{m} \sum_{j=1}^m 1\{x^{(j)} = x^{(i)}\} \\
 &= H(\hat{P}) - \frac{1}{m} \sum_{i=1}^m \log P_\theta(x^{(i)})
 \end{aligned}$$

Hence,

$$\arg \min_{\theta} D_{\text{KL}}(\hat{P} \| P_{\theta}) = \arg \max_{\theta} \sum_{i=1}^m \log P_{\theta}(x^{(i)})$$

Problem 3

(a)

$$\begin{aligned} \mathbb{E}_{y \sim p(y; \theta)} [\nabla_{\theta'} \log p(y; \theta') |_{\theta' = \theta}] &= \int_{-\infty}^{\infty} p(y; \theta) \nabla_{\theta'} \log p(y; \theta') |_{\theta' = \theta} dy \\ &= \int_{-\infty}^{\infty} \nabla_{\theta'} p(y; \theta') |_{\theta' = \theta} dy \\ &= \nabla_{\theta'} \int_{-\infty}^{\infty} p(y; \theta') dy \Big|_{\theta' = \theta} \\ &= \nabla_{\theta'} 1 |_{\theta' = \theta} \\ &= 0 \end{aligned}$$

(b) Let

$$\mu = \mathbb{E}_{y \sim p(y; \theta)} [\nabla_{\theta'} \log p(y; \theta') |_{\theta' = \theta}] = 0$$

Then,

$$\begin{aligned} \mathcal{I}(\theta) &= \text{Cov}_{y \sim p(y; \theta)} [\nabla_{\theta'} \log p(y; \theta') |_{\theta' = \theta}] \\ &= \mathbb{E}_{y \sim p(y; \theta)} [(\nabla_{\theta'} \log p(y; \theta') - \mu)(\nabla_{\theta'} \log p(y; \theta') - \mu)^T |_{\theta' = \theta}], \\ &= \mathbb{E}_{y \sim p(y; \theta)} [\nabla_{\theta'} \log p(y; \theta') \nabla_{\theta'} \log p(y; \theta')^T |_{\theta' = \theta}] \end{aligned}$$

(c)

$$\begin{aligned} \nabla_{\theta'} \log p(y; \theta') &= \frac{1}{p} \nabla_{\theta'} p(y; \theta') \\ \nabla_{\theta'}^2 \log p(y; \theta') &= -\frac{1}{p^2} \nabla_{\theta'} p(y; \theta') (\nabla_{\theta'} p(y; \theta'))^T + \frac{1}{p} \nabla_{\theta'}^2 p(y; \theta') \\ \mathbb{E}_{y \sim p(y; \theta)} [-\nabla_{\theta'}^2 \log p(y; \theta') |_{\theta' = \theta}] &= \int_{-\infty}^{\infty} -p \nabla_{\theta'}^2 \log p(y; \theta') |_{\theta' = \theta} dy \\ &= \int_{-\infty}^{\infty} \frac{1}{p} \nabla_{\theta'} p(y; \theta') (\nabla_{\theta'} p(y; \theta'))^T - \nabla_{\theta'}^2 p(y; \theta') \Big|_{\theta' = \theta} dy \\ &= \int_{-\infty}^{\infty} p \nabla_{\theta'} \log p(y; \theta') (\nabla_{\theta'} \log p(y; \theta'))^T \Big|_{\theta' = \theta} dy + \nabla_{\theta'}^2 \int_{-\infty}^{\infty} p(y; \theta') dy |_{\theta' = \theta} \\ &= \mathbb{E}_{y \sim p(y; \theta)} [\nabla_{\theta'} \log p(y; \theta') \nabla_{\theta'} \log p(y; \theta')^T |_{\theta' = \theta}] \\ &= \mathcal{I}(\theta) \end{aligned}$$

(d) Let $\theta + d = \tilde{\theta}$, $f(\tilde{\theta}) = D_{\text{KL}}(p_{\theta} \| p_{\tilde{\theta}})$.

$$\begin{aligned} f(\theta) &= D_{\text{KL}}(p_{\theta} \| p_{\theta}) = 0 \\ \nabla_{\theta'} f(\theta') &= \nabla_{\theta'} D_{\text{KL}}(p_{\theta} \| p_{\theta'}) \\ &= \nabla_{\theta'} \int_{-\infty}^{\infty} p_{\theta}(x) \log \frac{p_{\theta}(x)}{p_{\theta'}(x)} dx \\ &= -\nabla_{\theta'} \int_{-\infty}^{\infty} p_{\theta}(x) \log p_{\theta'}(x) dx \\ &= -\mathbb{E}_{x \sim p(x)} [\nabla_{\theta'} \log p_{\theta'}(x)] \end{aligned}$$

$$\begin{aligned}\nabla_{\theta'}^2 f(\theta') &= -\nabla_{\theta'} \int_{-\infty}^{\infty} p_{\theta}(x) \nabla_{\theta'} \log p_{\theta'}(x) dx \\ &= \mathbb{E}_{x \sim p(x)} [-\nabla_{\theta'}^2 \log p_{\theta'}(x)]\end{aligned}$$

The Taylor series of $f(\tilde{\theta})$ near θ is

$$\begin{aligned}f(\tilde{\theta}) &\approx f(\theta) + (\tilde{\theta} - \theta)^T \nabla_{\theta'} f(\theta')|_{\theta'=\theta} + \frac{1}{2}(\tilde{\theta} - \theta)^T (\nabla_{\theta'}^2 f(\theta')|_{\theta'=\theta}) (\tilde{\theta} - \theta) \\ &= 0 - d^T \mathbb{E}_{x \sim p(x)} [\nabla_{\theta'} \log p_{\theta'}(x)|_{\theta'=\theta}] + \frac{1}{2} d^T \mathbb{E}_{x \sim p(x)} [-\nabla_{\theta'}^2 \log p_{\theta'}(x)|_{\theta'=\theta}] d \\ &= \frac{1}{2} d^T \mathcal{I}(\theta) d\end{aligned}$$

(e)

$$\begin{aligned}\mathcal{L}(d, \lambda) &= \ell(\theta + d) - \lambda [D_{\text{KL}}(p_{\theta} \| p_{\theta+d}) - c] \\ &\approx \ell(\theta) + d^T \nabla_{\theta'} \ell(\theta')|_{\theta'=\theta} - \lambda \left[\frac{1}{2} d^T \mathcal{I}(\theta) d - c \right] \\ \nabla_d \mathcal{L}(d, \lambda) &= \nabla_{\theta'} \ell(\theta')|_{\theta'=\theta} - \lambda \mathcal{I}^T(\theta) d = \nabla_{\theta'} \ell(\theta')|_{\theta'=\theta} - \lambda \mathcal{I}(\theta) d \\ \nabla_{\lambda} \mathcal{L}(d, \lambda) &= \frac{1}{2} d^T \mathcal{I}(\theta) d - c\end{aligned}$$

Hence, we have system of linear equations:

$$\nabla_{\theta'} \ell(\theta')|_{\theta'=\theta} - \lambda \mathcal{I}^T(\theta) d = 0 \quad (\text{a})$$

$$\frac{1}{2} d^T \mathcal{I}(\theta) d - c = 0 \quad (\text{b})$$

From (a) we have

$$d = \frac{1}{\lambda} \mathcal{I}^{-1}(\theta) \nabla_{\theta'} \ell(\theta')|_{\theta'=\theta}$$

Plug it into (b):

$$\begin{aligned}\frac{1}{\lambda^2} (\nabla_{\theta'} \ell(\theta')|_{\theta'=\theta})^T \mathcal{I}^{-1}(\theta) \mathcal{I}(\theta) \mathcal{I}^{-1}(\theta) \nabla_{\theta'} \ell(\theta')|_{\theta'=\theta} &= 2c \\ \frac{1}{\lambda^2} (\nabla_{\theta'} \ell(\theta')|_{\theta'=\theta})^T \mathcal{I}^{-1}(\theta) \nabla_{\theta'} \ell(\theta')|_{\theta'=\theta} &= 2c \\ \lambda &= \sqrt{\frac{1}{2c} (\nabla_{\theta'} \ell(\theta')|_{\theta'=\theta})^T \mathcal{I}^{-1}(\theta) \nabla_{\theta'} \ell(\theta')|_{\theta'=\theta}}\end{aligned}$$

Plug it back into (a), which yields natural gradient:

$$d^* = \sqrt{\frac{2c}{(\nabla_{\theta'} \ell(\theta')|_{\theta'=\theta})^T \mathcal{I}^{-1}(\theta) \nabla_{\theta'} \ell(\theta')|_{\theta'=\theta}}} \mathcal{I}^{-1}(\theta) \nabla_{\theta'} \ell(\theta')|_{\theta'=\theta}$$

(f)

$$\begin{aligned}
\mathcal{I}(\theta) &= \mathbb{E}_{y \sim p(y; \theta)} [-\nabla_{\theta'}^2 \log p(y; \theta')|_{\theta'=\theta}] \\
&= \frac{1}{m} \sum_{i=1}^m \mathbb{E}_{y^{(i)} \sim p(y; \theta)} [-\nabla_{\theta'}^2 \log p(y^{(i)}; \theta')|_{\theta'=\theta}] \\
&= \frac{1}{m} \mathbb{E}_{y^{(i)} \sim p(y; \theta)} [-\nabla_{\theta'}^2 \sum_{i=1}^m \log p(y^{(i)}; \theta')|_{\theta'=\theta}] \\
&= \frac{1}{m} \mathbb{E}_{y \sim p(y; \theta)} [-\nabla_{\theta'}^2 \ell(\theta')|_{\theta'=\theta}] \\
&= \frac{1}{m} \mathbb{E}_{y \sim p(y; \theta)} [-\nabla_{\theta'}^2 \ell(\theta') \nabla_{\theta'}^2 \ell(\theta')|_{\theta'=\theta}] \nabla_{\theta'}^2 \ell(\theta')|_{\theta'=\theta} \\
&= \frac{1}{m} \mathbb{E}_{y \sim p(y; \theta)} [-1] \nabla_{\theta'}^2 \ell(\theta')|_{\theta'=\theta} \\
&= -\frac{1}{m} \nabla_{\theta'}^2 \ell(\theta')|_{\theta'=\theta}
\end{aligned}$$

The natural gradient is

$$\begin{aligned}
\tilde{d} &= \frac{1}{\lambda} \mathcal{I}^{-1}(\theta) \nabla_{\theta'} \ell(\theta')|_{\theta'=\theta} \\
&= -\frac{1}{\lambda m} (\nabla_{\theta'}^2 \ell(\theta'))^{-1} \nabla_{\theta'} \ell(\theta')|_{\theta'=\theta}
\end{aligned}$$

Newton's method has update rule

$$\begin{aligned}
\theta &:= \theta - (\nabla_{\theta'}^2 \ell(\theta')|_{\theta'=\theta})^{-1} \nabla_{\theta'} \ell(\theta')|_{\theta'=\theta} \\
&= \theta + \lambda m \tilde{d}
\end{aligned}$$

Problem 4

(a)

$$\begin{aligned}
\ell_{\text{semi-sup}}(\theta^{(t+1)}) &= \left(\sum_{i=1}^m \log \sum_{z^{(i)}} p(x^{(i)}, z^{(i)}; \theta^{(t+1)}) \right) + \alpha \sum_{i=1}^{\tilde{m}} \log p(\tilde{x}^{(i)}, \tilde{z}^{(i)}; \theta^{(t+1)}) \\
&\geq \left(\sum_{i=1}^m \sum_{z^{(i)}} Q_i^{(t)}(z^{(i)}) \log \frac{p(x^{(i)}, z^{(i)}; \theta^{(t+1)})}{Q_i^{(t)}(z^{(i)})} \right) + \alpha \sum_{i=1}^{\tilde{m}} \log p(\tilde{x}^{(i)}, \tilde{z}^{(i)}; \theta^{(t+1)}) \\
&\geq \left(\sum_{i=1}^m \sum_{z^{(i)}} Q_i^{(t)}(z^{(i)}) \log \frac{p(x^{(i)}, z^{(i)}; \theta^{(t)})}{Q_i^{(t)}(z^{(i)})} \right) + \alpha \sum_{i=1}^{\tilde{m}} \log p(\tilde{x}^{(i)}, \tilde{z}^{(i)}; \theta^{(t)}) \\
&= \ell_{\text{semi-sup}}(\theta^{(t)})
\end{aligned}$$

(b)

$$\begin{aligned}
Q^{(t)}(z) &:= p(z \mid x; \mu^{(t)}, \Sigma^{(t)}, \phi^{(t)}) = \frac{p(x \mid z; \mu^{(t)}, \Sigma^{(t)}, \phi^{(t)}) p(z; \mu^{(t)}, \Sigma^{(t)}, \phi^{(t)})}{p(x; \mu^{(t)}, \Sigma^{(t)}, \phi^{(t)})} \\
&= \frac{\frac{1}{(2\pi)^{d/2} |\Sigma_z^{(t)}|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu_z^{(t)})^T (\Sigma_z^{(t)})^{-1} (x - \mu_z^{(t)}) \right) \cdot \phi_z^{(t)}}{\sum_z \phi_z^{(t)} \frac{1}{(2\pi)^{d/2} |\Sigma_z^{(t)}|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu_z^{(t)})^T (\Sigma_z^{(t)})^{-1} (x - \mu_z^{(t)}) \right)}
\end{aligned}$$

(c) Consider

$$\begin{aligned} J(\mu, \Sigma, \phi) &= \left(\sum_{i=1}^m \sum_{z^{(i)}} Q_i^{(t)}(z^{(i)}) \log \frac{p(x^{(i)}, z^{(i)}; \mu, \Sigma, \phi)}{Q_i^{(t)}(z^{(i)})} \right) + \alpha \sum_{i=1}^{\tilde{m}} \log p(\tilde{x}^{(i)}, \tilde{z}^{(i)}; \mu, \Sigma, \phi) \\ &= \left(\sum_{i=1}^m \sum_{z^{(i)}} Q_i^{(t)}(z^{(i)}) \log \frac{\phi_{z^{(i)}} \mathcal{N}(x^{(i)}; \mu_{z^{(i)}}, \Sigma_{z^{(i)}})}{Q_i^{(t)}(z^{(i)})} \right) + \alpha \sum_{i=1}^{\tilde{m}} \log \left(\phi_{\tilde{z}^{(i)}} \mathcal{N}(\tilde{x}^{(i)}; \mu_{\tilde{z}^{(i)}}, \Sigma_{\tilde{z}^{(i)}}) \right) \end{aligned}$$

Update for μ :

$$\begin{aligned} \nabla_{\mu_j} J(\mu, \Sigma, \phi) &= \left(\sum_{i=1}^m Q_i^{(t)}(j) (\Sigma_j^{(t)})^{-T} (x^{(i)} - \mu_j) \right) + \alpha \sum_{i=1}^{\tilde{m}} \mathbf{1}\{\tilde{z}^{(i)} = j\} (\Sigma_j^{(t)})^{-T} (\tilde{x}^{(i)} - \mu_j) \\ &= \left(\sum_{i=1}^m Q_i^{(t)}(j) (\Sigma_j^{(t)})^{-1} x^{(i)} \right) + \alpha \sum_{i=1}^{\tilde{m}} \mathbf{1}\{\tilde{z}^{(i)} = j\} (\Sigma_j^{(t)})^{-1} \tilde{x}^{(i)} \\ &\quad - \left[\left(\sum_{i=1}^m Q_i^{(t)}(j) (\Sigma_j^{(t)})^{-1} \mu_j \right) + \alpha \sum_{i=1}^{\tilde{m}} \mathbf{1}\{\tilde{z}^{(i)} = j\} (\Sigma_j^{(t)})^{-1} \mu_j \right] \end{aligned}$$

Thus, if letting $\nabla_{\mu_j} J(\mu, \Sigma, \phi) = 0$, we have

$$\begin{aligned} &\left(\sum_{i=1}^m Q_i^{(t)}(j) (\Sigma_j^{(t)})^{-1} x^{(i)} \right) + \alpha \sum_{i=1}^{\tilde{m}} \mathbf{1}\{\tilde{z}^{(i)} = j\} (\Sigma_j^{(t)})^{-1} \tilde{x}^{(i)} \\ &= \left[\left(\sum_{i=1}^m Q_i^{(t)}(j) (\Sigma_j^{(t)})^{-1} \mu_j \right) + \alpha \sum_{i=1}^{\tilde{m}} \mathbf{1}\{\tilde{z}^{(i)} = j\} (\Sigma_j^{(t)})^{-1} \mu_j \right] \\ \mu_j &= \frac{\left(\sum_{i=1}^m Q_i^{(t)}(j) x^{(i)} \right) + \alpha \sum_{i=1}^{\tilde{m}} \mathbf{1}\{\tilde{z}^{(i)} = j\} \tilde{x}^{(i)}}{\left(\sum_{i=1}^m Q_i^{(t)}(j) \right) + \alpha \sum_{i=1}^{\tilde{m}} \mathbf{1}\{\tilde{z}^{(i)} = j\}} \end{aligned}$$

Update for ϕ :

$$\nabla_{\phi_j} J(\mu, \Sigma, \phi) = \left(\sum_{i=1}^m Q_i^{(t)}(j) \frac{1}{\phi_j} \right) + \alpha \sum_{i=1}^{\tilde{m}} \mathbf{1}\{\tilde{z}^{(i)} = j\} \frac{1}{\phi_j}$$

Note the constraint that $\sum_j \phi_j = 1$. With Lagrange multiplier we have equation

$$\begin{aligned} \nabla_{\phi_j} J(\mu, \Sigma, \phi) - \lambda &= 0 \\ \lambda \phi_j &= \left(\sum_{i=1}^m Q_i^{(t)}(j) \right) + \alpha \sum_{i=1}^{\tilde{m}} \mathbf{1}\{\tilde{z}^{(i)} = j\} \\ \lambda &= \sum_{j=1}^k \left(\sum_{i=1}^m Q_i^{(t)}(j) \right) + \alpha \sum_{i=1}^{\tilde{m}} \mathbf{1}\{\tilde{z}^{(i)} = j\} \\ &= \left(\sum_{i=1}^m \sum_{j=1}^k Q_i^{(t)}(j) \right) + \alpha \sum_{i=1}^{\tilde{m}} \sum_{j=1}^k \mathbf{1}\{\tilde{z}^{(i)} = j\} \\ &= m + \alpha \tilde{m} \end{aligned}$$

Hence,

$$\phi_j = \frac{\left(\sum_{i=1}^m Q_i^{(t)}(j) \right) + \alpha \sum_{i=1}^{\tilde{m}} \mathbf{1}\{\tilde{z}^{(i)} = j\}}{m + \alpha \tilde{m}}$$

Update for Σ :

$$\begin{aligned}
\nabla_{\Sigma_j} J(\mu, \Sigma, \phi) &= -\frac{1}{2} \left(\sum_{i=1}^m Q_i^{(t)}(j) \Sigma_j^{-1} \right) \\
&\quad + \frac{1}{2} \left(\sum_{i=1}^m Q_i^{(t)}(j) \Sigma_j^{-1} (x^{(i)} - \mu_j^{(t)}) (x^{(i)} - \mu_j^{(t)})^T \Sigma_j^{-1} \right) \\
&\quad - \frac{\alpha}{2} \left(\sum_{i=1}^{\tilde{m}} \mathbf{1}\{\tilde{z}^{(i)} = j\} \Sigma_j^{-1} \right) \\
&\quad + \frac{\alpha}{2} \sum_{i=1}^{\tilde{m}} \mathbf{1}\{\tilde{z}^{(i)} = j\} \Sigma_j^{-1} (\tilde{x}^{(i)} - \mu_j^{(t)}) (\tilde{x}^{(i)} - \mu_j^{(t)})^T \Sigma_j^{-1} \\
&= \frac{1}{2} \left(\sum_{i=1}^m Q_i^{(t)}(j) \Sigma_j^{-1} \left((x^{(i)} - \mu_j^{(t)}) (x^{(i)} - \mu_j^{(t)})^T \Sigma_j^{-1} - I \right) \right) \\
&\quad + \frac{\alpha}{2} \sum_{i=1}^{\tilde{m}} \mathbf{1}\{\tilde{z}^{(i)} = j\} \Sigma_j^{-1} \left((\tilde{x}^{(i)} - \mu_j^{(t)}) (\tilde{x}^{(i)} - \mu_j^{(t)})^T \Sigma_j^{-1} - I \right)
\end{aligned}$$

Let $\nabla_{\Sigma_j} J(\mu, \Sigma, \phi) = 0$, then

$$\begin{aligned}
0 &= \left(\sum_{i=1}^m Q_i^{(t)}(j) \left((x^{(i)} - \mu_j^{(t)}) (x^{(i)} - \mu_j^{(t)})^T \Sigma_j^{-1} - I \right) \right) \\
&\quad + \alpha \sum_{i=1}^{\tilde{m}} \mathbf{1}\{\tilde{z}^{(i)} = j\} \left((\tilde{x}^{(i)} - \mu_j^{(t)}) (\tilde{x}^{(i)} - \mu_j^{(t)})^T \Sigma_j^{-1} - I \right) \\
\left(\sum_{i=1}^m Q_i^{(t)}(j) \right) + \alpha \sum_{i=1}^{\tilde{m}} \mathbf{1}\{\tilde{z}^{(i)} = j\} &= \left(\sum_{i=1}^m Q_i^{(t)}(j) (x^{(i)} - \mu_j^{(t)}) (x^{(i)} - \mu_j^{(t)})^T \Sigma_j^{-1} \right) \\
&\quad + \alpha \sum_{i=1}^{\tilde{m}} \mathbf{1}\{\tilde{z}^{(i)} = j\} (\tilde{x}^{(i)} - \mu_j^{(t)}) (\tilde{x}^{(i)} - \mu_j^{(t)})^T \Sigma_j^{-1} \\
\Sigma_j &= \frac{\left(\sum_{i=1}^m Q_i^{(t)}(j) (x^{(i)} - \mu_j^{(t)}) (x^{(i)} - \mu_j^{(t)})^T \right) + \alpha \sum_{i=1}^{\tilde{m}} \mathbf{1}\{\tilde{z}^{(i)} = j\} (\tilde{x}^{(i)} - \mu_j^{(t)}) (\tilde{x}^{(i)} - \mu_j^{(t)})^T}{\left(\sum_{i=1}^m Q_i^{(t)}(j) \right) + \alpha \sum_{i=1}^{\tilde{m}} \mathbf{1}\{\tilde{z}^{(i)} = j\}}
\end{aligned}$$

(d) See Figure 1, Figure 2, and Figure 3.

(e) See Figure 4, Figure 5, and Figure 6.

(f) (i) Number of iterations to converge:

The semi-supervised EM converged in fewer iterations than the unsupervised EM. This is because the labeled examples provide additional information that helps guide the parameter updates toward the correct solution.

(ii) Stability:

The semi-supervised EM was more stable across different random initializations. The cluster assignments from different trials were more consistent, while the unsupervised EM produced more variations due to its dependence on the initial parameters.

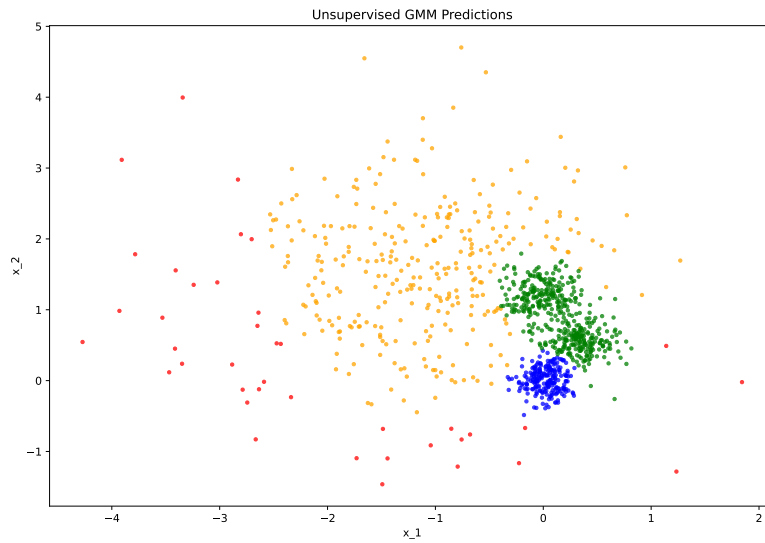


Figure 1: Unsupervised GMM predictions, trial 1.

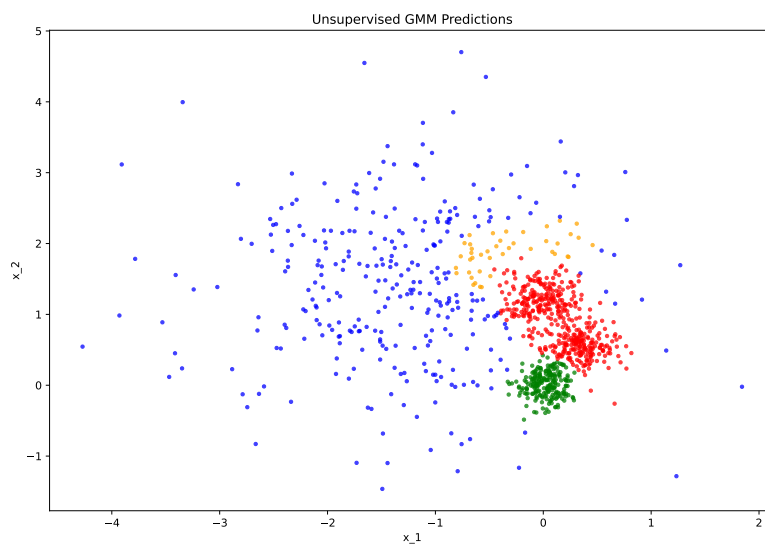


Figure 2: Unsupervised GMM predictions, trial 2.

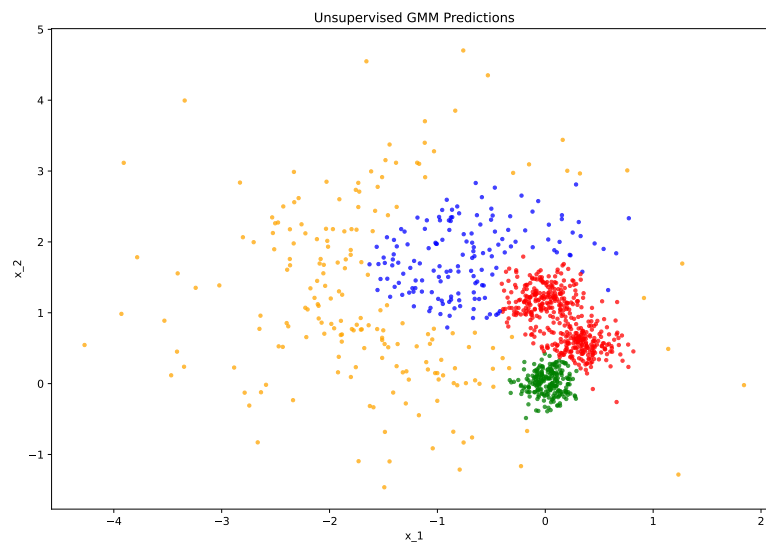


Figure 3: Unsupervised GMM predictions, trial 3.

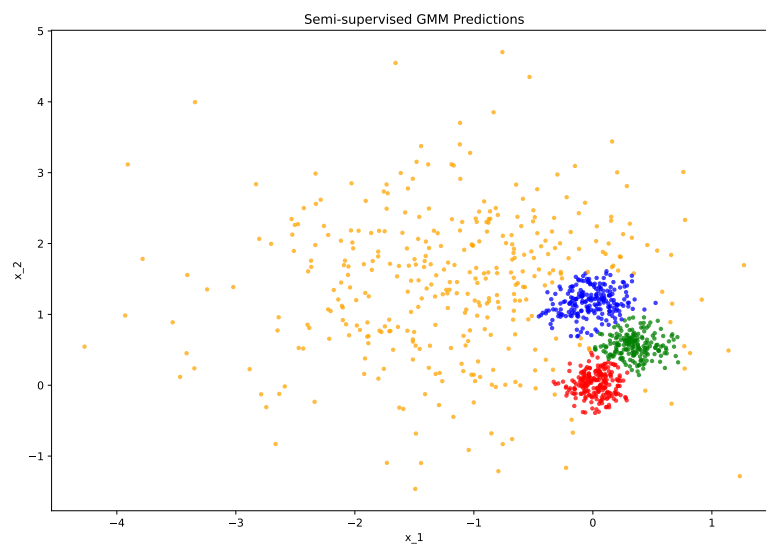


Figure 4: Semi-supervised GMM predictions, trial 1.

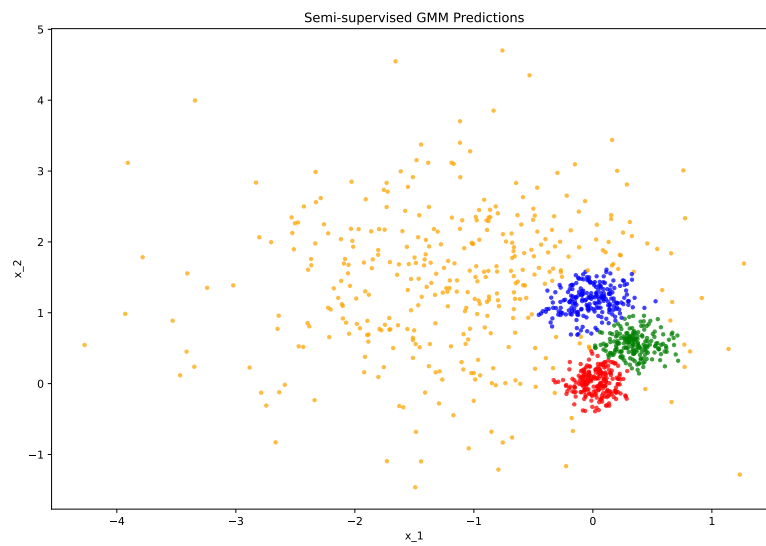


Figure 5: Semi-supervised GMM predictions, trial 2.

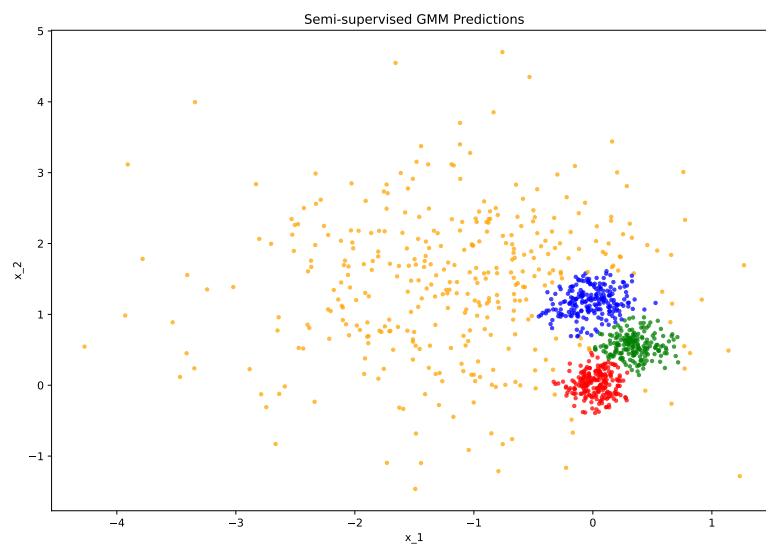


Figure 6: Semi-supervised GMM predictions, trial 3.

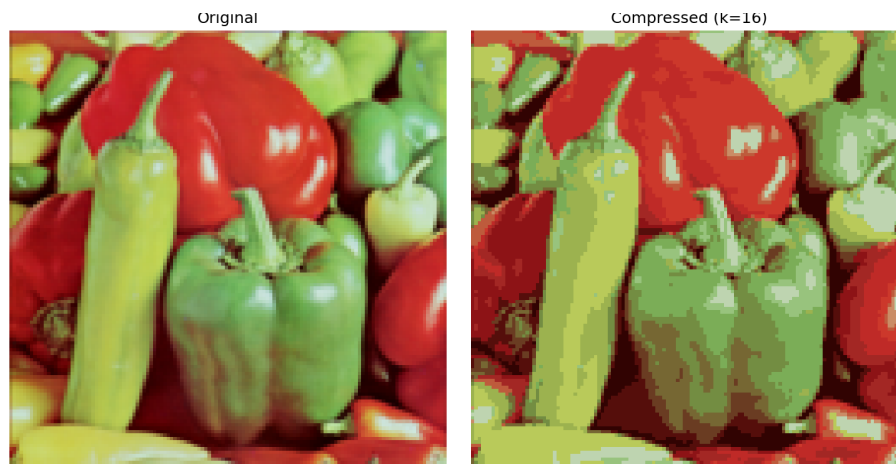


Figure 7: Comparison of original and compressed images

(iii) Overall quality of assignments:

The semi-supervised EM generally produced better cluster assignments. The labeled examples helped each Gaussian component match the correct cluster, especially for the low-variance Gaussian distributions. The unsupervised EM sometimes confused clusters or converged to different local optima, although it was still able to identify the main structure of the data.

Problem 5

(a) See Figure 7: Comparison of original and compressed images.

Code:

```
1 from matplotlib.image import imread, imsave
2 import matplotlib.pyplot as plt
3 import numpy as np
4
5 def k_means(k, x):
6     """
7     K-Means
8
9     Args:
10         k: number of clusters
11         x: m*m pixels image with r, g, and b channels, of shape (m, n, 3)
12
13     Returns:
14         Compressed image's np array of shape (m, n, 3)
15     """
16
17     min_iter = 30
18     max_iter = 1000
19     eps = 1e-3
20
21     m, n, _ = x.shape
```

```
22     x = x.astype(np.float64)
23
24     flat = x.reshape(-1, 3)
25     idx = np.random.choice(flat.shape[0], k, replace=False)
26     mu = flat[idx]
27     z = np.zeros((m, n), dtype=int)
28
29     it = 0
30     prev_mu = None
31
32     while (it < min_iter) or (prev_mu is None or np.linalg.norm(mu - prev_mu)
33         > eps and it < max_iter):
34         diff = x[:, :, np.newaxis, :] - mu[np.newaxis, np.newaxis, :, :]
35         dist = np.linalg.norm(diff, axis=3) # shape (m, n, k)
36         z = np.argmin(dist, axis=2) # shape (m, n)
37
38         prev_mu = mu.copy()
39         for i in range(k):
40             mask = (z == i)
41             if np.any(mask):
42                 mu[i] = np.mean(x[mask], axis=0)
43
44         it += 1
45     print(f"Converge in {it} iterations.")
46
47     compressed = mu[z]
48     return np.clip(compressed, 0, 255).astype(np.uint8)
49
50 def main():
51     A = imread('PS3/data/peppers-small.tif')
52
53     k = 16
54     compressed = k_means(k, A)
55
56     fig, axes = plt.subplots(1, 2, figsize=(10, 5))
57     axes[0].imshow(A)
58     axes[0].set_title('Original')
59     axes[0].axis('off')
60
61     axes[1].imshow(compressed)
62     axes[1].set_title(f'Compressed (k={k})')
63     axes[1].axis('off')
64
65     plt.tight_layout()
66     plt.savefig('PS3/output/peppers_compressed_comparison.png', dpi=150)
67     plt.show()
68
69     imsave('PS3/output/peppers_compressed.png', compressed)
70
71 if __name__ == '__main__':
72     main()
```

- (b) The color now can be represented by a 4-bit number (0 to 15) whereas originally it takes three 8-bits number. Therefore the compression factor is approximately 6.